Junyu Ren

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Research Interests

Verified agentic reasoning: eliminating unsupported empirical claims structurally rather than statistically — kernel-checked derivations, tool-attested evidence, fail-closed abstention, auditable proof obligations.
Mathematical foundations for AI: complexity limits of attention, topology of network expressivity, mechanistic interpretability of routing and composition.
AI for research-level mathematics: AI-(co)-mathematician workflows for long-horizon, proof-intensive problems, under strict proof-status discipline and adversarial review.

Education

University of Chicago 2024 – Present

PhD in Computational and Applied Mathematics.
Advisor: Lek-Heng Lim.
GPA: 4.0/4.0.

University of Oxford 2023 – 2024

MMathPhil, Master of Mathematics and Philosophy.
Thesis: *Koopman Operator Theory and Network-informed Dynamic Mode Decomposition.*
Supervisor: Renaud Lambiotte.
Grades: 74/100; cohort ranking 4/18.

University of Oxford 2020 – 2023

BA, Mathematics and Philosophy.
Grades: 77/100; cohort ranking 1/20.

Research and Publications

* *Equal contribution / co-first authorship; co-first authors listed in alphabetical order.*

Published and Accepted

- **Low-dimensional Topology of Deep Neural Networks.**
  Junyu Ren, Lek-Heng Lim.
  ICML 2026, Poster.
- **EG-VAR: A Path Toward Eliminating LLM Hallucination in Empirical Inference via Tool-Attested Kernel Proofs.**
  Junyu Ren.
  Accepted at three ICML 2026 workshops — three parts of one agenda:
  FAGEN — proof-checker feedback for agentic workflows.
  TAIGR — formal sidecars as technical governance for high-stakes outputs.
  PhilML ("From Prompts to Proof Obligations") — world-directed reasoning as auditable formal commitments.
- **When Do Transformer Components Compose? Validating a Log-Pool Decomposition Criterion.**
  Junyu Ren, Su Hyeong Lee, Risi Kondor.
  ICML 2026 Workshop PhilML, Poster.
- **MAPPA: Scaling Multiagent Systems with Process Rewards.**
  Ed Li*, **Junyu Ren***, Cat Yan.
  ICLR 2026 Workshop RSI.
- **Agent-as-a-Coach: Towards Fully Agentic, Stateful, and Tool-Augmented Process Rewards.**
  Ed Li*, **Junyu Ren***, Cat Yan, Kerem Goksel.
  ICLR 2026 Workshop MALGAI.
- **Build Your Personalized Research Group: A Multiagent Framework for Continual and Interactive Science Automation.**
  Ed Li*, **Junyu Ren***, Xintian Pan, Cat Yan, Chuanhao Li, Dirk Bergemann, Zhuoran Yang.
  arXiv:2510.15624, Oct. 2025.

- **Time-aware Feature Selection: Adaptive Temporal Masking for Stable Sparse Autoencoder Training.**
  Ed Li*, **Junyu Ren***.
  ICLR 2025 Workshop XAI4Science.
- **Unlocking Hierarchical Concept Discovery in Language Models Through Geometric Regularization.**
  Ed Li*, **Junyu Ren***.
  ICLR 2025 Workshop BuildingTrust.

Under Review and In Preparation

- **Training a 2-token attention is NP-hard.**
  Junyu Ren, Zehua Lai, Lek-Heng Lim.
  Manuscript under review, 2026.
- **Manifold-Weighted Neural Networks.**
  Junyu Ren, Lek-Heng Lim.
  Manuscript under review, 2026.
- **Optimization Dynamics Imprint Semantic Specificity in Contrastive Embedding Norms.**
  Ziwei Su, **Junyu Ren**, Victor Veitch.
  Manuscript under review, 2026.
- **Routing-Mediated Structural Pseudo-Alignment in Sparse Mixture-of-Experts.**
  Junyu Ren, Su Hyeong Lee, Risi Kondor.
  Manuscript under review (workshop), 2026.
- **Extensions of the Pierce–Birkhoff conjecture beyond hyperplane partitions.**
  Junyu Ren, Zehua Lai, Lek-Heng Lim.
  Manuscript in preparation, 2026.
- **Hodge-decomposed Hochschild cohomology of socle quotient rings.**
  Junyu Ren, Lek-Heng Lim, Ke Ye.
  Manuscript in preparation, 2026.

Awards and Honors

McCormick Fellowship, University of Chicago	2024
Gibbs Prize for FHS Mathematics and Philosophy, University of Oxford	2023
Oriel College Scholarship, University of Oxford	2020 – 2024

Talks & Presentations

- *Low-dimensional topology of deep neural networks.*
  AMS 2026 Spring Eastern Sectional Meeting, Special Session, Boston College, Mar. 2026.
- *Hallucination as a Type Error.*
  IDEAL Workshop on Language Models and Text Elicitation Mechanisms, University of Illinois Chicago, May 2026.

Professional Services

Reviewer, ICML 2026 *(Gold Reviewer Reward)*	2026
Reviewer, ICML 2026 Workshops	2026
Referee, Communications Physics (Nature Portfolio)	2026
Referee, The Journal of Geometric Analysis	Jun. 2026 – present
Referee, Chaos, Solitons & Fractals	Jun. 2026 – present
Referee, Linear Algebra and Its Applications	May 2025 – present
Referee, SIAM Journal on Computing	Oct. 2025 – present
Referee, SIAM Journal on Matrix Analysis and Applications	Oct. 2025 – present
Teaching Assistant, University of Chicago	Oct. 2025 – present